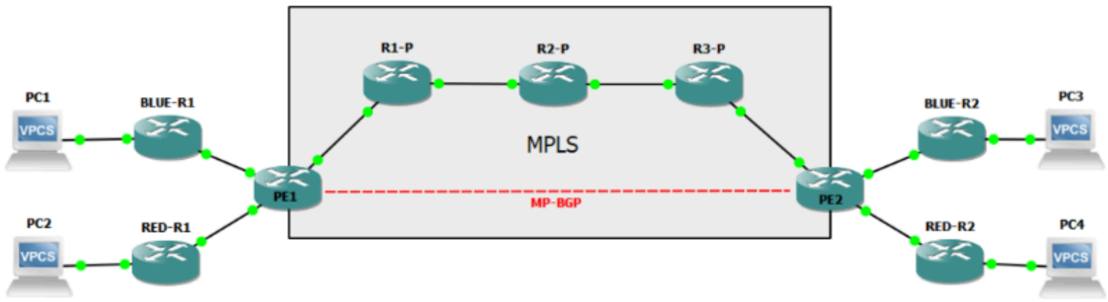
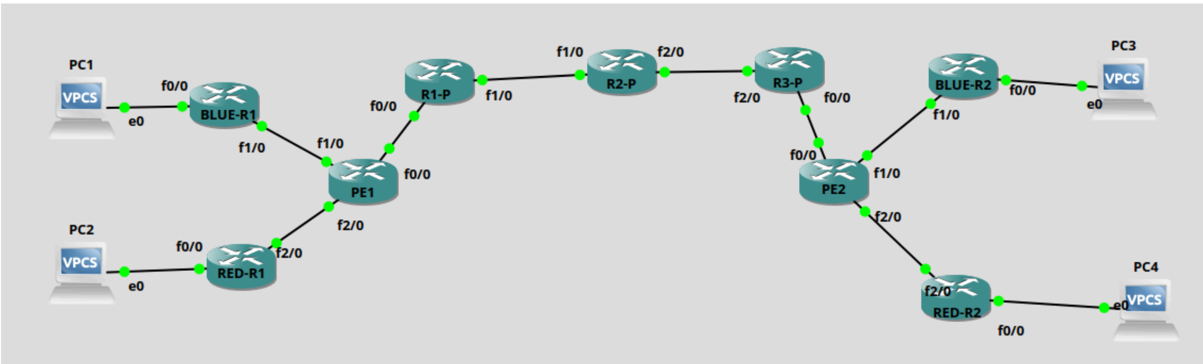
TP2 - R302 MPLS

On réalise ce schéma :



On obtiens :



On configure nos routeurs :

PE1 :

```
en
conf t
int f0/0
ip address 10.10.12.1 255.255.255.252
no sh
int l0
ip address 1.1.1.1 255.255.255.255
no sh
```

R1-P :

```
en
conf t
int f0/0
ip address 10.10.12.2 255.255.255.252
no sh
int f1/0
ip address 10.10.23.1 255.255.255.252
no sh
int l0
ip address 2.2.2.2 255.255.255.255
no sh
```

R2-P :

```
en
conf t
int f1/0
ip address 10.10.23.2 255.255.255.252
no sh
int f2/0
ip address 10.10.34.1 255.255.255.252
no sh
int l0
ip address 3.3.3.3 255.255.255.255
no sh
```

R3-P :

```
en
conf t
int f2/0
ip address 10.10.34.2 255.255.255.252
no sh
int f0/0
ip address 10.10.45.1 255.255.255.252
no sh
int l0
ip address 4.4.4.4 255.255.255.255
no sh
```

PE2 :

```
en
conf t
int f0/0
ip address 10.10.45.2 255.255.255.252
no sh
```

```
int l0
ip address 5.5.5.5 255.255.255.255
no sh
```

On configure le routage ospf :

PE1 :

```
en
conf t
router ospf 1
router-id 1.1.1.1
int l0
ip ospf 1 area 0
int f0/0
ip ospf 1 area 0
```

R1-P :

```
en
conf t
router ospf 1
router-id 2.2.2.2
int l0
ip ospf 1 area 0
int f0/0
ip ospf 1 area 0
int f1/0
ip ospf 1 area 0
```

R2-P :

```
en
conf t
router ospf 1
router-id 3.3.3.3
int l0
ip ospf 1 area 0
int f1/0
ip ospf 1 area 0
int f2/0
ip ospf 1 area 0
```

R3-P :

```
en
conf t
router ospf 1
router-id 4.4.4.4
int l0
ip ospf 1 area 0
int f0/0
ip ospf 1 area 0
int f2/0
ip ospf 1 area 0
```

PE2 :

```
en
conf t
router ospf 1
router-id 5.5.5.5
int l0
ip ospf 1 area 0
int f0/0
ip ospf 1 area 0
```

On configure le MPLS et le LDP :

PE1 :

```
en
conf t
mpls label protocol ldp
mpls label range 100 199
mpls ldp router-id l0 force
mpls ip

int f0/0
mpls ipVotre identifiant
```

R1-P :

```
en
conf t
mpls label protocol ldp
mpls label range 200 299
mpls ldp router-id l0 force
mpls ip

int f0/0
mpls ip
```

```
int f1/0
mpls ip
```

R2-P :

```
en
conf t
mpls label protocol ldp
mpls label range 300 399
mpls ldp router-id l0 force
mpls ip

int f1/0
mpls ip

int f2/0
mpls ip
```

R3-P :

```
en
conf t
mpls label protocol ldp
mpls label range 400 499
mpls ldp router-id l0 force
mpls ip

int f0/0Votre identifiant
mpls ip

int f2/0
mpls ip
```

PE2 :

```
en
conf t
mpls label protocol ldp
mpls label range 500 599
mpls ldp router-id l0 force
mpls ip

int f0/0
mpls ip
```

On vérifie qu'ils connaissent les labels de leurs voisins :

```
show mpls ldp bindings
```

On vérifie que les labels sont utilisés :

```
traceroute 10.10.12.2
```

Ou :

```
traceroute 10.10.45.2
```

Pour voir la table CEF on fait la commande :

```
show ip CEF
```

On met en place le VRF :

Sur PE1 :

```
en
conf t
ip vrf BLUE
rd 100:100
route-target export 100:1000
route-target import 100:1000

int f1/0
ip vrf forwarding BLUE
ip address 10.0.1.1 255.255.255.0
no sh

ip vrf RED
rd 200:200
route-target export 200:2000
route-target import 200:2000

int f2/0
ip vrf forwarding RED
ip address 10.0.2.1 255.255.255.0
no sh
```

Sur PE2 :

```
en
conf t
ip vrf BLUE
rd 100:100
route-target export 100:1000
route-target import 100:1000

int f1/0
ip vrf forwarding BLUE
ip address 10.0.3.1 255.255.255.0
no sh

ip vrf RED
rd 200:200
route-target export 200:2000
route-target import 200:2000

int f2/0
ip vrf forwarding RED
ip address 10.0.4.1 255.255.255.0
no sh
```

Sur BLUE-R1 :

```
en
conf t
ip route 0.0.0.0 0.0.0.0 10.0.1.1
int f1/0
ip address 10.0.1.2 255.255.255.0
no sh
int f0/0
ip address 192.168.1.1 255.255.255.0
no sh
```

Sur RED-R1 :

```
en
conf t
ip route 0.0.0.0 0.0.0.0 10.0.2.1
int f2/0
ip address 10.0.2.2 255.255.255.0
no sh
int f0/0
ip address 192.168.3.1 255.255.255.0
no sh
```

Sur BLUE-R2 :

```
en
conf t
ip route 0.0.0.0 0.0.0.0 10.0.3.1
int f1/0
ip address 10.0.3.2 255.255.255.0
no sh
int f0/0
ip address 192.168.2.1 255.255.255.0
no sh
```

Sur RED-R2 :

```
en
conf t
ip route 0.0.0.0 0.0.0.0 10.0.4.1
int f2/0
ip address 10.0.4.2 255.255.255.0
no sh
int f0/0
ip address 192.168.4.1 255.255.255.0
no sh
```

PC1 :

```
ip 192.168.1.2 255.255.255.0 192.168.1.1
```

PC2 :

```
ip 192.168.3.2 255.255.255.0 192.168.3.1
```

PC3 :

```
ip 192.168.2.2 255.255.255.0 192.168.2.1
```

PC4 :

```
ip 192.168.4.2 255.255.255.0 192.168.4.1
```

Mise en place du BGP :

PE1 :

```
router bgp 1
bgp log-neighbor-changes
neighbor 5.5.5.5 remote-as 1
neighbor 5.5.5.5 update-source Loopback0

address-family vpnv4
neighbor 5.5.5.5 activate
neighbor 5.5.5.5 send-community extended
exit-address-family

address-family ipv4 vrf BLUE
redistribute connected
redistribute static

exit-address-family

address-family ipv4 vrf RED
redistribute connected
redistribute static
exit-address-family
```

PE2 :

```
router bgp 1
bgp log-neighbor-changes
neighbor 1.1.1.1 remote-as 1
neighbor 1.1.1.1 update-source Loopback0

address-family vpnv4
neighbor 1.1.1.1 activate
neighbor 1.1.1.1 send-community extended
exit-address-family

address-family ipv4 vrf BLUE
redistribute connected
redistribute static
exit-address-family

address-family ipv4 vrf RED
redistribute connected
redistribute static
exit-address-family
```

Configuration_VRF_TP2.pdf

Resume.pdf

c3640-jk9s-mz.124-16(1).bin